Jiajie Jin

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About me

- I am a Ph.D. student at the Gaoling School of Artificial Intelligence, Renmin University of China, advised by Prof. Zhicheng Dou (expected graduation in 2029). I received my M.S. from the same school and my B.S. in Statistics from the University of Science and Technology of China.
- My research works toward the **next generation of LLM agents** — which, to take on real-world research, must be **autonomous** over long horizons, **efficient**, and **trustworthy**. I contribute to each of these along three threads:
  - **Long-horizon autonomy:** agents that plan, search, reason, and run long-horizon experiments on their own (*WebThinker*, *HiRA*, *FinSight*, *Arbor*).
  - **Efficient knowledge use (effectiveness & efficiency):** infrastructure and methods that let LLMs exploit external knowledge accurately at a fraction of the cost (*FlashRAG*, *LongRefiner*, *BIDER*).
  - **Trustworthiness (factuality & traceability):** making agent reasoning verifiable and auditable, so that numbers are reproducible and conclusions are traceable to evidence (*VeriGraph*).
- I am especially driven by turning these capabilities into systems and products that people can use.

Educational Experience

- **Gaoling School of Artificial Intelligence, Renmin University of China (RUC)** *Sep. 2025 – Present*
  Ph.D in Artificial Intelligence
  Advisor: Prof. Zhicheng Dou
- **Gaoling School of Artificial Intelligence, Renmin University of China (RUC)** *Sep 2023 – July 2025*
  M.S in Artificial Intelligence
  Advisor: Prof. Zhicheng Dou
- **University of Science and Technology of China (USTC)** *Sep 2019 – Jul 2023*
  B.S. in Statistics

Publications (First / Co-first Author)

FinSight: Towards Real-World Financial Deep Research
***J Jin**, Y Zhang, Y Xu, H Qian, Y Zhu, Z Dou*
ACL 2026 Main, Oral, SAC Highlight (CCF-A)

LaSER: Internalizing Explicit Reasoning into Latent Space for Dense Retrieval
***J Jin**, Y Zhang, M Li, D Long, P Xie, Y Zhu, Z Dou*
Annual ACM SIGIR Conference on Research and Development in Information Retrieval 2026 (CCF-A)

HiRA: A Hierarchical Reasoning Framework for Decoupled Planning and Execution in Deep Search
***J Jin**, X Li, G Dong, Y Zhang, Y Zhu, Y Zhao, H Qian, Z Dou*
Annual ACM SIGIR Conference on Research and Development in Information Retrieval 2026 (CCF-A)

Hierarchical Document Refinement for Long-Context Retrieval-Augmented Generation
***J Jin**, X Li, G Dong, Y Zhang, Y Zhu, Y Wu, Z Li, Q Ye, Z Dou*
Annual Meeting of the Association for Computational Linguistics 2025 Main Conference, Oral (CCF-A)

FlashRAG: A Modular Toolkit for Efficient Retrieval-Augmented Generation Research
***J Jin**, Y Zhu, Z Dou, G Dong, X Yang, C Zhang, T Zhao, Z Yang, JR Wen*
The Web Conference 2025 Resource Track, Oral (CCF-A)

WebThinker: Empowering Large Reasoning Models with Deep Research Capability
X Li, **J Jin***, G Dong*, H Qian, Y Zhu, Y Wu, JR Wen, Z Dou*
NeurIPS 2025 (CCF-A); Hugging Face Daily Papers #1

Bider: Bridging Knowledge Inconsistency for Efficient Retrieval-Augmented LLMs via Key Supporting Evidence
***J Jin**, Y Zhu, Y Zhou, Z Dou*

Findings of the Association for Computational Linguistics: ACL 2024 (CCF-A)

Preprints / Under Submission

VeriGraph: Towards Verifiable Data Analysis Agents
J Jin*, *Z Yang**, W Liao, Y Hu, G Dong, X Li, Y Zhu, Z Dou*
Preprint, 2026

Toward Generalist Autonomous Research via Hypothesis-Tree Refinement (Arbor)
J Jin*, *Y Hu**, K Qiu, Q Dai, C Luo, G Dong, X Li, T Zhao, X Ma, G Zhang, Z Wu, B Liu, Z Yang, L Li, L Wang, H Qian, Y Zhu, Z Dou*
Preprint / Technical Report, 2026; BAAI Conference 2026 AI4Science Highlight Paper; Hugging Face Daily Papers #1

Projects

FlashRAG: A Python Toolkit for Efficient RAG Research github.com/FlashRAG

- Modular RAG framework with 20+ algorithm reproductions and 40+ datasets; independently built most of the codebase.
- **3.5k+** stars (top-3 on GitHub trending); datasets downloaded **100k+** times and cited **200+** times.

Arbor: Toward Generalist Autonomous Research via Hypothesis-Tree Refinement github.com/Arbor

- General autonomous-research framework that turns scattered trials into a durable research state via **Hypothesis-Tree Refinement** for long-horizon experimentation and self-improvement.
- Best held-out results on all **6** tasks (**2.5**× the average gain of Codex and Claude Code) and **86.36%** Any Medal on MLE-Bench Lite. Built during my internship at Microsoft Research Asia.

WebThinker: Empowering Large Reasoning Models with Deep Research Capability github.com/WebThinker

- Teach large reasoning models to think deeply while autonomously searching the web, enabling o1/R1-style open-ended deep research; **1.4k+** GitHub stars.

FinSight: Towards Real-World Financial Deep Research github.com/FinSight

- Multi-agent financial deep research system automating the full pipeline from multi-source data collection to **publication-ready reports** with professional charts and evidence tracing; **1st place (1/1289)** in AFAC2025.

Internships

- **Microsoft Research Asia** *Mar. 2026 – Present*
  Topic: Auto Research
  Mentor: Kai Qiu, Chong Luo
- **Alibaba, Tongyi Lab** *Sep. 2025 – Mar. 2026*
  Topic: Latent Reasoning for Foundation Embedding Model
  Mentor: Dingkun Long

Awards

- SAC Highlight, ACL 2026 (for FinSight) 2026
- Dean's Scholarship (16 University-Wide), Renmin University of China Jun 2026
- Qingyuan InnoVibe 2026 – Most-Watched Academic Rising Star, BAAI Conference Jun 2026
- First Prize (1/1289 teams), AFAC Financial Intelligence Innovation Competition Aug 2025
- First-Class Academic Scholarship, Renmin University of China Dec 2024
- WeiLai (JAC-NIO) Special Scholarship, USTC Dec 2022
- First Prize (Anhui Province), National College Student Mathematics Competition 2021